

Initial Investigations into Information Retention and Perception on Virtual Human Race

Deyrel Diaz *

Samaneh Zamanifard †

Andrew T. Duchowski‡

Matias Volonte §

School of Computing, Clemson University

ABSTRACT

Virtual human avatars have long been studied in the field of embodied conversational agents. Most studies have focused on understanding how verbal and non-verbal cues build long-term relationships, trust, and credibility. Some studies have even gone as far as looking into characteristics like clothes, accessories, and race to see what effects they may have on the interlocutor. We seek feedback on an investigative study where we look to better understand how avatar race affects the perception of the avatar, information retention, and eye gaze behavior. We discuss the technical design and research methodology.

Keywords: Virtual Human, Eye-Tracking, Area of Interest, Augmented Reality, Race.

Index Terms: Human-centered computing—Visualization—Empirical studies in visualization

1 INTRODUCTION

In the realm of interactive virtual human(s) (VH), customization of physical attributes is gaining traction for creating relatable and intuitive embodied conversational agents [5, 6, 8]. This customization is crucial as racial/ethnic concordance impacts user satisfaction and perception, evident in healthcare and educational settings [2, 7]. Previous studies have highlighted that individuals engage more positively with virtual entities that mirror their racial background. Leary et. al found that culturally tailoring religious black virtual agents for predominately black church communities, through participatory design, to discuss health promotion had positive impacts on participants' self-efficacy, connection to the avatar, and spiritual dialogue [3]. Moreover, the "Other-race effect" underlines a cognitive bias where individuals better recognize faces of their own race, influencing social interactions and visual attention [4]. These studies suggest that racial concordance might play a significant role in enhancing educational engagement and effectiveness. Our study delves into this phenomenon within augmented reality, analyzing how VH' racial diversity affects users' visual attention and learning outcomes.

This research seeks to investigate how the racial diversity of virtual agents influences user engagement and information retention outcomes. Employing a within-subjects design, participants engaged with digital avatars representing a range of demographics, including Asian, African-American, Caucasian, Hispanic, and Middle Eastern. These virtual entities provide knowledge on emerging technology topics such as Artificial Intelligence, Gene Therapy, Nanotechnology, Robotics, and Cryptocurrency. Data on visual attention through eye-tracking of the facial area(s) of interest (AOI), correctly answering



Figure 1: Set of male and female avatars used for this experiment.

the questions about the topics and personal feedback is gathered. We present the current implementation and seek feedback on how to improve the methodology or enhance the experiment regarding embodied conversational VH.

2 EXPERIMENTAL DESIGN

For our experimental setup, we utilized the Microsoft HoloLens 2 (HL2) head-mounted display, chosen for its wireless functionality and advanced eye-tracking capabilities. This device is supported by a robust computing setup, featuring a high-end PC with an Intel Core i7 processor, NVIDIA GeForce RTX 2080 graphics card, and 32 GB of RAM. This combination ensures a seamless mixed reality experience with the capability to handle complex simulations.

The setup was intentionally simplistic. Positioned at approximately eye level, the main camera within the XR rig was placed 1.6 meters high and 1.2 meters ahead of the avatar. This placement strategically situated the avatar within the user's personal to-social space proxemic zone.

Eye-tracking is a pivotal component of our study, facilitated by the HL2's integrated system and further refined using the Microsoft Mixed Reality Toolkit (MRTK3) alongside the Extended Eye Tracking SDK. This setup ensures accurate tracking of both combined and individual eye gaze data, essential for analyzing user engagement and interaction within the augmented reality environment. Moreover, participants can use their eye tracking to select the answers to the given questions.

The augmented reality simulations were developed using Unity 3D, employing the MRTK for an immersive and interactive user experience. Selection and interaction within the simulation rely heavily on eye gaze, calibrated automatically for each user through MRTK3, ensuring a smooth and intuitive interface.

Our selection of VH was meticulously curated from the Rocket-box avatar collection, aiming to represent a broad spectrum of racial backgrounds. This selection process was guided by distinct facial characteristics such as shape, skin color, and features, to ensure a diverse yet consistent representation across the avatars used in the study.

To accurately measure visual attention, we attached capsule col-

*email:deyrelid@clemson.edu

†email:szamani@clemson.edu

‡email:duchowski@clemson.edu

§email:mvolont@clemson.edu

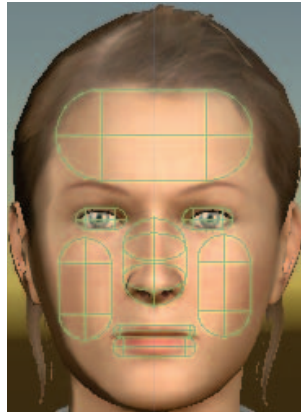


Figure 2: The areas of interest standardized on each virtual human.



Figure 3: The user interface shows a question during the demo that utilizes eye-tracking and turns green with gaze selection.

leaders using a uniquely developed tool to key areas of interest on the avatars, such as the eyes, nose, and mouth. These colliders are attached to the rig of the avatar and dynamically adjust with the avatars' movements, allowing us to precisely track gaze intersections and thereby assess user engagement and focus while the avatar is talking to them.

We are interested in not only how visual attention may change depending on the avatar demographics, but also relative to the participant's gender and race/ethnicity.

To achieve a realistic movement for the avatar, we positioned it in a casual standing pose. We employed the SALSA LipSync Suite (version 2.5.4) to set up nine visemes and four emotional expressions using the avatars' built-in blend shapes. To enhance the naturalism, we incorporated random eye movements, allowing the eyes to shift left or right with a 20% probability during each expression. Additionally, we configured the avatar's head to rotate along the x and z axes at the neck joint, enabling it to track the primary user's movement. To not induce uncanny effects of the voice of the avatar despite changing demographics. We randomized the pitch for avatars within each gender from 0.5 to 1.5.

3 PROCEDURE

In this experiment, participants begin by completing a consent form, pre-survey, and demographic questionnaire at a fixed desktop setup. They are then introduced to the study and instructed on using the HL2, including making selections by focusing on an item for over two seconds.

Participants are first presented with a demo avatar (not included in the selection of avatars shown in Figure 1) that welcomes them. This avatar allows the novelty effect of the VH and the HL2 to diminish before continuing with the experiment. Participants also get practice with using their eye gaze to select answers during this stage.

After the demo avatar disappears, participants are presented with a randomly assigned VH (dependent on the condition). The VH gives a brief two-minute lecture on a randomly assigned emerging technology topic mentioned previously. During the presentation, we will monitor eye movements to identify focal points on the avatar AOI (see Figure 2). Following the lecture, participants answer 14 multiple-choice questions. Four questions are specifically geared towards assessing attention to the material. The other 10 include a sanity check if they can correctly identify the avatar, and their feelings regarding trust, friendliness, likability, warmth, competence, similarity, and awareness.

This process repeats for each VH, with avatars reappearing in a

random sequence to prevent selection order. After interacting with all VHs, participants remove the HL2 and complete a post-survey on the desktop giving subjective feedback.

4 ANTICIPATED RESULTS AND DISCUSSION

The study will employ quantitative analysis techniques to assess the impact of VH demographics on user engagement and learning outcomes. Eye-tracking data will be analyzed to determine the duration and frequency of gaze on predefined areas of interest (AOIs) on the VH eight AOI. Statistical methods, such as ANOVA and regression analysis, will be used to explore the relationships between VH race, participant demographics, visual attention, and learning outcomes.

We look to filter the visual attention data through fixation and saccades best practices [1]. Simple rules of thumb can help reduce the incidence of incorrect detections.

It is anticipated that participants will exhibit varied visual attention and learning outcomes based on the racial concordance between themselves and the VH. Previous studies suggest that individuals may engage more positively with VH that mirrors their own racial background, potentially leading to better information retention and satisfaction.

By understanding how racial diversity or bias in VH affects user engagement and learning, developers can create more personalized and accessible learning experiences that cater to a diverse user base.

ACKNOWLEDGMENTS

This research is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. (#1744593). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

REFERENCES

- [1] A. T. Duchowski, N. A. Gehrer, M. Schöenberg, and K. Krejtz. Art facing science: Artistic heuristics for face detection: tracking gaze when looking at faces. In *Proceedings of the 11th ACM symposium on eye tracking research & applications*, pp. 1–5, 2019.
- [2] A. C. King, T. W. Bickmore, M. I. Campero, L. A. Pruitt, and J. L. Yin. Employing virtual advisors in preventive care for underserved communities: results from the compass study. *Journal of health communication*, 18(12):1449–1464, 2013.
- [3] T. K. O'Leary, E. Stowell, E. Kimani, D. Parmar, S. Olafsson, J. Hoffman, A. G. Parker, M. K. Paasche-Orlow, and T. Bickmore. Community-based cultural tailoring of virtual agents. In *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents*, pp. 1–8, 2020.
- [4] A. J. O'toole, K. A. Deffenbacher, D. Valentin, and H. Abdi. Structural aspects of face recognition and the other-race effect. *Memory & Cognition*, 22:208–224, 1994.
- [5] D. Parmar, S. Olafsson, D. Utami, and T. Bickmore. Looking the part: The effect of attire and setting on perceptions of a virtual health counselor. In *Proceedings of the 18th international conference on intelligent virtual agents*, pp. 301–306, 2018.
- [6] L. Ring, D. Utami, and T. Bickmore. The right agent for the job? the effects of agent visual appearance on task domain. In *Intelligent Virtual Agents: 14th International Conference, IVA 2014, Boston, MA, USA, August 27-29, 2014. Proceedings 14*, pp. 374–384. Springer, 2014.
- [7] P. R. Spence, K. A. Lachlan, S. A. Spates, and X. Lin. Intercultural differences in responses to health messages on social media from spokespeople with varying levels of ethnic identity. *Computers in Human Behavior*, 29(3):1255–1259, 2013.
- [8] M. Volante, S. V. Babu, H. Chaturvedi, N. Newsome, E. Ebrahimi, T. Roy, S. B. Daily, and T. Fasolino. Effects of virtual human appearance fidelity on emotion contagion in affective inter-personal simulations. *IEEE transactions on visualization and computer graphics*, 22(4):1326–1335, 2016.